16 A 100 g block of ice at −10.0 °C is placed into 150 g of water initially at 20.0 °C inside a perfectly insulated calorimeter. What is the final temperature of the mixture and the mass of ice remaining?

Specific heat capacity of water = 4180 J kg^{-1} K^{-1}

Specific latent heat of fusion of ice = 334000 J kg^{-1}

Specific heat capacity of ice = 2100 J kg^{-1} K^{-1}

A 0 K with 31.3 g of ice remaining

B 0 K with 68.7 g of ice remaining

C 0 °C with 31.3 g of ice remaining

D 0 °C with 68.7 g of ice remaining

Ans: D

Heat available from water cooling to 0°C:

$$Q_{water} = mc\Delta T = 0.150 \times 4180 \times 20.0 = 12540 \text{ J}$$

Heat needed to warm ice from −10.0°C to 0°C:

$$Q_{ice} = mc\Delta T = 0.100 \times 2100 \times 10.0 = 2100 \text{ J}$$

Remaining heat available to melt ice:

$$Q_{remaining} = 12540 - 2100 = 10440 \text{ J}$$

Mass of ice melted:

$$m = \frac{10440}{334000} = 0.03126 \text{ kg} = 31.3 \text{ g}$$

Ice remaining:

$$100 - 31.3 = 68.7 \text{ g}$$